

(12) [1]+[13]+[20] [1] [13] [20]

: 2026 3 19 : Exqutor

: , ,

1.3 [1] [13] [20]

1.1 [1] [13] [20]

[1] *AnalyticDB-V (VLDB 2020)*

AnalyticDB-V Alibaba 가 , (SQL) (,) **SQL** . :

1. **VAQ (Vector-Augmented Analytical Query)** : SQL , SELECT *
FROM items WHERE ANNS(embedding, query_vec, 10) AND price < 1000
가 1000 가
2. **VGPQ** : Product Quantization () ANN
3. **(Accuracy-Aware CBO)**: ANN ,

: . AnalyticDB-V **10%** 가 , 가 .
Exqutor .

[13] *pgvector (Open Source, 2021~[13])*

PostgreSQL pgvector 가 **DB** , , HNSW/IVFFlat , (L2, ,) . :

1. **SQL** : , , , , CTE,
2. : pg_dump, ,
3. **GitHub 12,000+** :

: pgvector **33.3%(= 1/3)** . #define DEFAULT_SEL 0.333333
, PostgreSQL , , ~ .

[20] *ICDE 2024 [20]: PostgreSQL [20] [20] [20]*

Purdue University **PostgreSQL** 가 가 . 5 가 :

1. (): HNSW 30~50% → Faiss 18
2. (): , MVCC 가 20~50% 가
3. (): 8KB , MVCC Faiss 8 → I/O
4. (): WAL , MVCC Faiss 96 (1,000 : PostgreSQL 48 vs Faiss 30)
5. (): 33.3% → 0.001% 33,300 , 425

: 1~3 PostgreSQL (, MVCC,) , 4~5 . **Exqutor** 5 ()
pgvector 1,000 .

1.2 如何選擇合適的數據結構

$$\vdots$$

[1] AnalyticDB-V (2020)

$$\vdash \text{SQL} \rightarrow \text{VAQ}$$

(10%)

[13] pgvector (2021~)

AnalyticDB-V

(33.3%)

→ 가

↓

[20] ICDE 2024 (2024)

└ pgvector

5가, vs

(33.3%)

**
 **

↓

Exqutor (2025)

$$\mathbb{L}^1[20]$$

ECQO ** **

→ pgvector 1,000

$$\vdots$$

1. **[1]** : “ + SQL ” ,
2. **[13]** : .AnalyticDB-V ,가
3. **[20]** : .pgvector 5 가 , 가
4. **Exqutor** : .[20] ,

2.3 如何選擇合適的數據結構：

2.1 질문: “데이터베이스가 무엇인가?”

“ ” .

AnalyticDB-V

pgvector

ICDE 2024

2.2 실험 데이터셋 구성: 실험 데이터셋 구성

실험 1: 실험 데이터셋 구성

: 1,000,000
: (< 0.5)

- :
 - A: 5 → 0.0005%
AnalyticDB-V : 10% → : 20,000
pgvector : 33.3% → : 66,600
 - B: 5,000 → 0.5%
AnalyticDB-V : 10% → : 20
pgvector : 33.3% → : 66
 - C: 500,000 → 50%
AnalyticDB-V : 10% → 5
pgvector : 33.3% → 1.5

실험 2: 실험 데이터셋 구성

: 10,000,000 (768 BERT)
:

- :
 - " ": 10,000 → 0.1%
pgvector : 33.3% → : 333
 - " ": 1,000,000 → 10%
pgvector : 33.3% → : 3.3

실험 3: ICDE 2024 실험 데이터셋 구성

: Wikipedia 10,000,000
: < 1.0 (가 가 100)

: 0.001% (100 / 10,000,000)
pgvector : 33.3%

: 33,300

- :
 - 개수 : 850
 - : 2
 - : 425

2.3 가계 부채의 위험 관리

PostgreSQL, , :

11

```
SELECT o.order_id, o.customer_id, SUM(l.price * l.quantity) AS total
FROM orders o
JOIN lineitem l ON o.order_id = l.order_id
WHERE l.product_embedding <-> query_vec < 0.5
    AND o.order_date > '2025-01-01'
    AND l.category = 'electronics'
GROUP BY o.order_id, o.customer_id;
```

□□□□: □□ □□□ = 0.1% (□□□ □□), □□ □□ □□ = 500 □

pgvector ():- lineitem : 600 - : 600 $\times$ 33.3% = 200 - : **Seq Scan on orders** (100) $\rightarrow$ lineitem 600

$$: - : 600 \times 0.1\% = 6,000 - : \quad \mathbf{6,000} \rightarrow \text{orders}$$

⋮

pgvector :

$$100 \times 600 = 600 \rightarrow$$
$$\vdots$$
$$6,000 \times 100 = 60 \rightarrow$$

: 1,000

2.4 時間と空間のギャップ(Gap) 問題

[1] AnalyticDB-V

[13] pgvector

[20] ICDE 2024

Exqutor : [20] “ , 가 ” → Exqutor ECQO

3. Problem Statement

3.1 数据 模型 模型

: (Vector Range Query) SQL , () .

$$\vdots$$
$$\vdots$$

SELECT ... FROM R
WHERE $\sigma_{\text{vec}}(\text{v_col}, \text{q_vec}, \epsilon)$ **AND** $\sigma_{\text{scalar}}(\text{scalar_col})$

:- R: $(|R| = n)$ - v_col: (d) - σ_{vec} : (ϵ) - σ_{scalar} :

:

sel_vec :

():

sel_vec = 0.333333 (pgvector) or 0.1 (AnalyticDB-V)

→ 가

():

sel_vec = $|\{r \in R : \text{distance}(r.\text{v_col}, \text{q_vec}) < \epsilon\}| / |R|$

→

:

가 가 :

:

$|\sigma_{\text{vec}} \cap \sigma_{\text{scalar}}| \approx n \times 0.333 \times \text{sel_scalar}$

→ , ,

:

$|\sigma_{\text{vec}} \cap \sigma_{\text{scalar}}| \approx n \times \text{actual_sel_vec} \times \text{sel_scalar}$

→

3.2 가 가 가 가 가 ?

2.2.1 가 가 가

AnalyticDB-V 10% :- , 20

pgvector 33.3% :- : 0.001% ~ 90% - 33,300

2.2.2 가 가 가 가

DB (, MCV):

WHERE age > 30 **AND** salary < 100000

PostgreSQL :- age > 30 : → ~90% - salary < 100000 : → ~90% - :

가

가 :

WHERE embedding <-> query_vec < 0.5

1. 가 : 768

- 1D : 10~100

- 768D : = 10^{768} ()
- 2. :
 - 1 0.5 가 0.1%
 - 2 0.5 가 50%
 -

3. :

2.2.3

: “ ?”

: - (1~2ms) - - “ ”

Exqutor : 가가가, 1~2ms (1,000)가

3.3

3.3.1

:
WHERE age > 30
: ()

:
WHERE embedding <-> query_vec < 0.5
: q_vec
, q_vec →

3.3.2

(Curse of Dimensionality):

- 1 : d = (2d)
- 2 : d = (πd^2)
- 3 : d = ($(4/3)\pi d^3$)
- 768 : d " " = d^{768}

:
→
→

3.3.3

HNSW/IVFFlat :

HNSW :
- :

- ef_search 가
- ef_search=100 → recall 95%
- ef_search=500 → recall 99.9%

:

ef_search 가스 가 ?

99% ?
95% ?

3.3.4 인덱스 인덱스 인덱스

:

SELECT ... FROM R WHERE vec_condition AND scalar_condition

: K

↓

(가)

↓

: K'

→ K ≠ K'

가 ?

3.4 Exqutor 인덱스 인덱스 인덱스

3.4.1 ECQO(Effective Cardinality-aware Query Optimization)인덱스 인덱스

: “가 , 1~2ms ”

:

1. : WHERE embedding <-> query_vec < ε
2. HNSW (1~2ms)
→ K
3. PostgreSQL K
- 4.

:

가 ()

3.4.2 인덱스-인덱스 인덱스

: 1~2ms ()

: 1,000 ()

: 100ms → 100μs

- : 1ms → 100ms (100)

3.4.3 PostgreSQL 벡터 쿼리

pgvector :

1. : PostgreSQL (planner hook)
 - pgvector
 -
 2. :
 - (distance < 0.5)
 - Top-K (ORDER BY distance LIMIT 10)
 -
 3. : PostgreSQL 가
 - pgvector (MySQL, SQLite)
-

4. Exqutor 벡터 쿼리 실행

4.1 벡터 쿼리 Exqutor 실행 벡터 쿼리

예 1: [1] *AnalyticDB-V = VAQ* 벡터 쿼리, 벡터 쿼리

Exqutor : Related Work

: - VAQ(Vector-Augmented Analytical Query) - SQL (Alibaba) - ()

Exqutor : - AnalyticDB-V: (vs) - Exqutor: (,)

:

Exqutor :
"AnalyticDB-V SQL
,
."

예 2: [13] *pgvector* = 벡터 쿼리, 33.3% 벡터 쿼리

Exqutor :

: - 가 DB - 33.3% - 가 (12,000+ GitHub = =)

Exqutor : 1. : DB pgvector 가 2. **baseline**: 33.3% 3. : PostgreSQL

:

Exqutor :

pgvector:
- : 33.3%
- : 850 ()

Exqutor (ECQO):

- :
- : 0.85
- : 1,000

:

Exqutor :

"pgvector PostgreSQL ,
SQL .
33.3%
,
.
pgvector ECQO
."

3: [20] ICDE 2024 = pgvector , ,

Exqutor :

: - pgvector 5 가 - vs - → Exqutor

Exqutor :

ICDE 2024 :

"pgvector 5 가 :
1~3 (, ,):
→ (PostgreSQL)
→

4 ():
→
→ , WAL 가

5 ():
→
→ , ,
"

Exqutor : 5

:

Exqutor :
 "Zhang et al. (ICDE 2024) PostgreSQL
 5가 .
 가 ,
 . ECQO
 ."

4.2 Exqutor 가 vs vs

: 5 () → 1,000

ECQO :

```
# (1~2ms 가 )
def estimate_vector_cardinality(query_vector, threshold):
    # pgvector HNSW
    results = hnsw_index.range_search(query_vector, threshold)
    return len(results) #

# PostgreSQL
def planner_hook(parse, ...)
    if has_vector_condition(parse):
        true_cardinality = estimate_vector_cardinality(...)
        set_selectivity(true_cardinality)
    return optimized_plan

:
```

0.001%

0.5%

50%

: 1~4

1: ()

: HNSW →

Exqutor : 가

: VBASE(Vector-Buffer Architecture) 가

2: ()

: , MVCC 가

Exqutor : 가

: immutable MVCC

3: ()

: 8KB , MVCC
Exqutor : 가
: 가

4: ()

: WAL , MVCC , maintenance_work_mem
Exqutor : 가
: , WAL ()

4.3 Exqutor 실행 환경 구축

Exqutor 가 :

1. (20): Faiss
2. (8):
3. : 가

: - pgvector : Faiss 20,000 () - Exqutor : Faiss 20 ()

: "Exqutor pgvector , PostgreSQL "

5. 실험 환경 (15)

5.1 실험 (2)

" .

Alibaba AnalyticDB-V(VLDB 2020), DB pgvector, ICDE 2024
PostgreSQL .

, , , . Exqutor 가 .
 , , Exqutor 가 ."

5.2 AnalyticDB-V 실험 (3)

" AnalyticDB-V .

AnalyticDB-V Alibaba 가 2020 VLDB . : (SQL) (,) SQL
 .

, : 1. (Faiss) → 1,000 2. ID SQL → SELECT * FROM products
WHERE product_id IN (...) 3. 가,

AnalyticDB-V :

```
SELECT * FROM products
WHERE ANNS(image_embedding, query_img, 10) -- top-10
AND price < 100000 -- SQL
```

SQL 가 .

AnalyticDB-V 가 SQL . 가 ‘ , 가 ’ .

가 : ?

AnalyticDB-V . “ 10% 가 ” . , 100 10 .

. 1 , 100 , 50 . ‘ 10% ’ 가 .

AnalyticDB-V ‘ + SQL ’ , ’ ’ . 가 Exqutor ."

5.3 pgvector (3)

" pgvector .

pgvector AnalyticDB-V PostgreSQL . 2021 12,000 GitHub DB .

pgvector 가 SQL . (, ,), HNSW IVFFlat , PostgreSQL SQL (, ,) .

pgvector 가 : 33.3% .

:

#define DEFAULT_SEL 0.333333

“ 33.3%가?” , . IVF 1/3 ’ .

가 . ICDE 2024 .

Wikipedia 1,000 :

: 0.001% (100 1) pgvector : 33.3% : 33,300

?

가 33.3% ’ . 0.001% , .

:

(pgvector): 850

(): 2

: 425

Exqutor 가 .
pgvector , ~ . "

5.4 ICDE 2024 : : (3)

"가 ICDE 2024 . 'PostgreSQL 가 ?' .

Purdue University pgvector , 5 가 .

1: ()

PostgreSQL . (HNSW) , (Latch) .
 , /가 30~50% . Faiss , pgvector 18 .

2: ()

PostgreSQL : 1. 2. 3. (xmin, xmax, cmin, cmax) 4. MVCC 가 (가?) 5.

. 20~50% .

3: ()

PostgreSQL HNSW 8KB . , .
:- Faiss: 150MB (+) - PostgreSQL: 1,200MB (, 8)
 , 100 .

4: ()

1,000 PostgreSQL :- PostgreSQL: 48 가 - Faiss: 30

96 . WAL (), MVCC , .

5: () ←

가 . pgvector 33.3% . 0.001%~90% , .
:

5 가 :- 1~3 (PostgreSQL) - , MVCC, PostgreSQL - PostgreSQL
• 4~5
- , WAL
-

"4~5 " , '가 ' . 가 Exqutor 가 .

5 :- pgvector : 1,000 - pgvector + 1~4: 20,000

1~4 , 5() . 5 1,000 → 20 (1~4)."

5.5 **Exqutor** (2)

"

[1] **AnalyticDB-V (2020)** - : SQL - : VAQ , - :

[13] **pgvector (2021~)** - : (33.3%) - : 12,000+ ,

[20] **ICDE 2024 (2024)** - : 5 가 , vs - : (5) - : “가”

Exqutor : 5

: 1. 2. HNSW (1~2ms) 3. 4. PostgreSQL 5.

- : - 가 : 1~2ms - : 1,000 () - : 100ms → 100μs

Exqutor , ."

5.6 **Exqutor** (1)

" Exqutor 가 ?

ECQO(Effective Cardinality-aware Query Optimization) .

:

1. PostgreSQL

2. (embedding <-> query_vec < ε)

3. HNSW

→ K

4. PostgreSQL selectivity = K / |R|

5.

: - pgvector () - (, Top-K) - PostgreSQL

: - TPC-H : 1,000 - -SQL : 100 - : 1~2ms ()

, [20] 1~4 , , 가 ."

5.7 **Exqutor** (1)

"

:

1. **ECQO**

- (, Top-K,)
 -
 -
- 2.
- TPC-H
 - (Wikipedia, MS COCO, GloVe)
 -
- 3.
- - 1~4
 - PostgreSQL
- 4.
- 1~4 Exqutor
 - ECQO
 - pgvector
- : - pgvector - PostgreSQL - pgvector
- ?"

6. 検索/分析 実行

6.1 検索 実行

/

Exqutor

6.2 検索 実行 実行

AnalyticDB-V

pgvector

Exqutor

6.3 ☐ ☐ ☐

HNSW

6.4 ☐ ☐ ☐ ☐ ☐

(A)

(B)

(C)

Wikipedia

7. ☐ Q&A (☐ ☐, 15+ ☐)

☐ 1: “☐ ☐ ☐ ☐ ☐ ☐?”

:

→ → → .

[1] AnalyticDB-V “ SQL ” . (10% 가).

[13] pgvector , 33.3% 가 . , .

[20] ICDE 2024 pgvector , 5 가 . 까 “ , ” .

Exqutor [20]가 “ ” .

☐ 2: “☐ ☐ ☐ ☐ ☐?”

:

Exqutor 까 가 :

1. : AnalyticDB-V

– “ +SQL ”

– Alibaba

2. : pgvector

- 가 (12,000+)
- 33.3% 가
-

3. : ICDE 2024

-
-
- “가 ”

Exqutor 가 “ pgvector ” . “ -SQL ” .

□□ 3: “□□□□□ □□ □ □□□ □□□□?”

:

PostgreSQL , , .

:

:

SELECT * FROM orders o
JOIN lineitem l **ON** o.order_id = l.order_id
WHERE l.embedding <-> query_vec < 0.5

1: 0.1%, pgvector 33.3% - : “가 orders → lineitem ” - : 가 embedding
 → orders - :

2: 50%, pgvector 33.3% - : “가 가 embedding ” - : 가 - : 5~10

, .

□□ 4: “□□□ 33.3%가 □ □□□□ □□ □□□□□□”

:

pgvector embedding <-> query_vec < 0.5 1/3 .

:

1,000,000 - 0.5 : 100 - pgvector 가 가 333,333

: - “333,333 , ” - 1,000,000

: - 100 - 10,000

: 1,000 “ 300 가 ≡ , 가 ” . 1 300 .

: - pgvector (): Faiss 20,000 () - pgvector + Exqutor: Faiss 20 () - : 1,000 (5)

Exqutor : “ 1,000 , PostgreSQL (, MVCC,) . ”

□□ 7: “□□ □□□ □ □□□□□ □□?”

:

1 : ECQO pgvector - - (, Top-K,) - -

2 : 가 - : - Wikipedia (1,000 , 2048D) - MS COCO (123 , 2048D) - GloVe (119 , 300D) - SIFT (1.28 , 128D)

- :
 - (baseline)
 - + SQL ()
 - + + ()
 - TPC-H
- :
 - 가
 -
 -
 -

3 : - ECQO - - 1~4 - PostgreSQL

: - TPC-H : 100~1,000 - : 50~500 - : 1~3ms ()

□□ 8: “□□ □□□□ □□ □□□ □□□?”

:

: - pgvector: 12,000+ GitHub , - “pgvector 가 ” - “ ”, “ ”

Exqutor :

- 가
 - PostgreSQL
 - pgvector
 - 가
- - 가 : 가 1~2ms
 - : 가 850 → 0.85

- : “1~2ms 가 1,000 ? ”
 - 3. **DB**
 - : “ DB(Milvus, Weaviate), pgvector”
 - Exqutor : “SQL pgvector + Exqutor”
 - : PostgreSQL 가 DB
 - 4.
 - Vector-Augmented Analytical Query(VAQ)
 -
 - DB
- : DB JSON . . .
-

9: “쿼리 쿼리결과 쿼리 쿼리 쿼리, 쿼리 쿼리 쿼리?”

- :
- , .
- :
- :
- : embedding <-> query_vec < 0.5
- HNSW 1
- : 1~2ms (100 , 10 ~100)
- :
- : ~10ms
- : ~20ms
- : ~50ms
- ECQO (): ~2ms
- : ~82ms vs ~80ms (가 2~3%)
- :
- (pgvector):
 - : 100ms
 - : 850
 - : 850.1
- (Exqutor):
 - : 102ms (가 2ms)
 - : 0.85
 - : 0.952
- :
- 850.1 / 0.952 = 893 (1,000)

- :

가 : 2ms
: 850 - 0.85 = 849.15
: 849.15 / 2ms = 424,575

, 2ms 849 . 가 .

□□ 10: “□□ □□ □□□□ □□ □□?”

:

.

:

1: SELECT * FROM images WHERE embedding <-> q1 < 0.5

2: SELECT * FROM images WHERE embedding <-> q2 < 0.5

...

N: SELECT * FROM images WHERE embedding <-> qN < 0.5

가↗가↘ :

1: HNSW (q1) → K1

2: HNSW (q2) → K2

...

N: HNSW (qN) → KN

HNSW : - HNSW () - - PostgreSQL

:

:

1: 2ms

2: 2ms

...

100: 2ms

: 200ms ()

: 1. 2. HNSW 3. ECQO

□□ 11: “HNSW □□□□ □□ □□□□ □□□□ □□ □□ □□ □□?”

:

. HNSW , .

HNSW ef_search :

```

ef_search = 100: recall ~95% ( 5% )
ef_search = 500: recall ~99% ( 1% )
ef_search = 1000: recall ~99.9% ( )

```

```

:

ECQO      (ef_search = ) .

// ECQO
results = hnsw_index.range_search(
    query_vector,
    threshold,
    ef_search = 1000 //
)
cardinality = len(results)

```

- :- ef_search : 1~2ms 가 (2~3ms) - : 99.9% - : <0.1% ()

```

:

ef_search ( , ):
- : 0.5ms
- : 5%
- : 5% → pgvector 33.3%

```

```

ef_search ( , ):
- : 2ms
- : <0.1%
- :

```

```

: ef_search
: 2ms 가 0.1% vs 5% 가

```

□□ 12: “가□□□□ (□□ □□ □□, □□ □□) □□□ □□□?”

```

:

```

```

.

```

```

:

```

```

SELECT * FROM items
WHERE embedding1 <-> q1 < 0.5
AND embedding2 <-> q2 < 0.3
AND price > 1000
AND category = 'electronics';

```

ECQO :

1. :

```

- embedding1 <-> q1 < 0.5 → K1 = 5,000
- embedding2 <-> q2 < 0.3 → K2 = 50,000
2.      :
- price > 1000 AND category = 'electronics' → K3 = 200,000 (    )
3.      :
- 가 :  $K1 \times K2 \times K3 / |R|^2$ 
-
4.      :
-
-
:
(TPC-H ):
pgvector ( ): 1,000
Exqutor ( ): 50
: 20
,      Exqutor      .

```

□□ 13: “pgvector 가□□ □ □□□ □?”

```

:
,가.      .
: PostgreSQL (Planner Hook)
PostgreSQL      (extension point)      :
// PostgreSQL
planner_hook = my_planner;

PlannedStmt *my_planner(
    Query *parse,
    const char *query_string,
    int cursorOptions,
    ParamListInfo boundParams
) {
    // 1.
    if (has_vector_condition(parse)) {
        // 2.
        estimate_vector_cardinality(parse);
    }

    // 3.
    PlannedStmt *plan = standard_planner(parse, ...);
}

```

```
    return plan;
}
```

: 1. pgvector 2. PostgreSQL 3. 4. pgvector

:

```
CREATE EXTENSION ecqo;
```

```
--
-- pgvector
```

.

□□ 14: “AnalyticDB-V □ 10% vs pgvector □ 33.3% 가 □ □ □ □ □ □ □?”

:

. ...

:

:

0.5% :
AnalyticDB-V: 10% → 20
pgvector: 33.3% → 66
→ pgvector 가

5% :
AnalyticDB-V: 10% → 2
pgvector: 33.3% → 6.7
→ pgvector 가

50% :
AnalyticDB-V: 10% → 5
pgvector: 33.3% → 1.5
→ AnalyticDB-V 가

가 : - AnalyticDB-V: Alibaba 가 - pgvector: 가 ,

: “ . . ”

□□ 15: “Exqutor □ □□ □□□ □□ □□□□?”

:

Exqutor :

1. :

```
SELECT * FROM a
JOIN b ON a.id = b.a_id
JOIN c ON b.id = c.b_id
WHERE a.embedding <-> q1 < 0.5
AND b.embedding <-> q2 < 0.3
AND c.embedding <-> q3 < 0.2
AND a.created_at > NOW() - INTERVAL '7 days'
AND ... ( )
```

: 가 →

2. (Data Drift):

: = 0.5% → A

: 가

: 2% → A

: → ICDE 2024

3. (Anomalous Distribution):

:

:

: 95% ()

pgvector (33.3%): - ()

Exqutor (95%): () →

→ Exqutor 가

4. :

HNSW :

: WHERE embedding <-> q < ε AND category = 'electronics'

: 10,000

category : 100

```
: Exqutor      ,      pgvector      .
```

$$\vdots$$

가 :

VLDB/SIGMOD :- 가 -

2. System Paper vs Techniques:

3. :

- $$\vdots$$

1. ICDE VLDB Workshop:

-

2. System & Performance Track:

- SIGMOD “System and Performance”

3. PostgreSQL :

- PGConf PGDay
- 가

⋮

주 :- : 1,000 - 가 : GitHub - : pgvector 12,000+

: 1. : () 2. : ICDE, VLDB Workshop, PGConf () 3. : PostgreSQL/pgvector ()

: Exqutor “ ” “ ” . , .

8. 발표 자료

8.1 발표 자료 소개

[1] AnalyticDB-V

[13] pgvector

[20] ICDE 2024

8.2 Exqutor 소개

Exqutor 소개 :

- [1] + [13] + [20] “ ”
- : ECQO
- : pgvector 1,000
- : PostgreSQL transparent integration

8.3 발표 자료 소개

1. “ ”
 - 가 , , 끼
 2. “1,000 1~2ms ”
 - -
 3. “Exqutor ”
 - 1~4 가 , 5
 - scope
 4. “pgvector 12,000+ ”
 - ,
-